

P-BLOCK MASTER STATEMENT

ASSIGNMENT

GROUP 17 - HALOGEN FAMILY

PART II — GROUP 17

Q28. EXCLUSIVE ORDER QUESTION - periodic properties [Count the correct statements.]

- (1) Atomic radius: $F < Cl < Br < I$.
- (2) Ionic radius: $F^- < Cl^- < Br^- < I^-$.
- (3) First ionisation enthalpy: $F > Cl > Br > I$.
- (4) Electronegativity: $F > Cl > Br > I$.
- (5) Electron gain enthalpy becomes less negative from Cl to I.
- (6) Magnitude of hydration enthalpy: $F^- > Cl^- > Br^- > I^-$.
- (7) Melting point: $F_2 < Cl_2 < Br_2 < I_2$.
- (8) Boiling point: $F_2 < Cl_2 < Br_2 < I_2$.
- (9) Bond length: $F-F < Cl-Cl < Br-Br < I-I$.
- (10) Bond dissociation enthalpy follows $Cl-Cl > Br-Br > F-F > I-I$ for the values.
- (11) Oxidising power in aqueous solution: $F_2 > Cl_2 > Br_2 > I_2$.
- (12) Reactivity of elemental halogens decreases down the group.
- (13) Acid strength of hydrogen halides: $HF < HCl < HBr < HI$.
- (14) H-X bond dissociation enthalpy: $H-F > H-Cl > H-Br > H-I$.
- (15) Electron gain enthalpy follows $F < Cl < Br < I$ in negativity, with fluorine the most negative.

Answer: _____

Q29. Oxidation states and chemical reactivity [Count the correct statements.]

- (1) All halogens exhibit the -1 oxidation state.
- (2) Chlorine, bromine and iodine also exhibit +1, +3, +5 and +7 oxidation states.
- (3) Higher positive oxidation states occur mainly in compounds with oxygen and fluorine.
- (4) The +4 and +6 oxidation states also occur in oxides and oxoacids of chlorine and bromine.
- (5) Fluorine has no d orbitals in its valence shell in the valence-shell model.
- (6) Fluorine cannot expand its octet in that model.
- (7) Because fluorine is the most electronegative element, it shows only -1 oxidation state in its compounds.
- (8) All halogens are highly reactive.

- (9) They react with metals and non-metals to form halides.
- (10) Elemental halogen reactivity decreases down the group.
- (11) Ready acceptance of an electron explains their oxidising nature.
- (12) F_2 is the strongest oxidising halogen.
- (13) Fluorine commonly shows +7 oxidation state in oxoacids.
- (14) Halogen reactivity increases from fluorine to iodine.
- (15) Iodine is the strongest oxidising halogen.

Answer: _____

Q30. Oxidising power, displacement and reactions with water [Count the correct statements.]

- (1) F_2 oxidises Cl^- , Br^- and I^- .
- (2) Cl_2 oxidises Br^- and I^- .
- (3) Br_2 oxidises I^- .
- (4) I_2 oxidises Br^- spontaneously in the reverse of the normal displacement trend.
- (5) A halogen generally oxidises halide ions of higher atomic number.
- (6) Fluorine oxidises water to oxygen.
- (7) The fluorine-water reaction produces H^+ , F^- and O_2 in ionic form.
- (8) Chlorine and bromine react with water to form the corresponding hydrogen halide and hypohalous acid.
- (9) The reaction of iodine with water is non-spontaneous in this discussion.
- (10) Iodide can be oxidised by oxygen in acidic medium.
- (11) Chlorine cannot oxidise iodide.
- (12) Bromine readily oxidises fluoride to fluorine.
- (13) The decreasing oxidising ability down the group is consistent with the electrode potentials.
- (14) The water reactions provide a direct chemical illustration of relative oxidising power.
- (15) F_2 reacts with water only to form HF; no oxygen is produced.

Answer: _____

Q31. Chlorine - complete uses coverage [Count the correct statements.]

- (1) Chlorine is used for bleaching wood pulp used in paper and rayon manufacture.
- (2) It is used for bleaching cotton and textiles.
- (3) It is used in extraction of gold and platinum.
- (4) It is used in manufacture of dyes.
- (5) It is used in manufacture of drugs.
- (6) It is used in manufacture of organic compounds such as CCl_4 and $CHCl_3$.
- (7) It is used in manufacture of DDT.
- (8) It is used in manufacture of refrigerants.
- (9) It is used for sterilising drinking water.
- (10) It is used in preparation of phosgene, $COCl_2$.
- (11) It is used in preparation of tear gas, CCl_3NO_2 .
- (12) It is used in preparation of mustard gas.
- (13) Chlorine is mainly a reducing agent for food preservation.
- (14) Chlorine is unsuitable for water sterilisation.
- (15) Its industrial importance combines bleaching, disinfection, metallurgy and chemical manufacture.

Answer: _____

Q32. Hydrogen chloride - uses [Count the correct statements.]

- (1) HCl is used in the manufacture of chlorine.
- (2) It is used in manufacture of ammonium chloride.
- (3) It is used in manufacture of glucose from corn starch.
- (4) It is used for extracting glue from bones.
- (5) It is used for purifying bone black.
- (6) It has medicinal uses.
- (7) It is used as a laboratory reagent.
- (8) HCl is as the principal raw material for manufacture of ozone.
- (9) HCl is stated to be unusable in laboratory chemistry because it is too weak.

- (10) Its use in chlorine manufacture links hydrogen chloride directly with halogen chemistry.
- (11) Its reaction with weaker-acid salts makes it a useful general laboratory acid.
- (12) HCl is a bleaching agent for wool and silk.
- (13) Hydrochloric acid is an important reagent in both inorganic preparation and analysis.
- (14) HCl is used as the oxidising component of the Contact process.
- (15) The uses include both industrial and laboratory applications.

Answer: _____

Q33. Advanced oxoacid acidity and oxidation-state challenge [Count the correct statements.]

- (1) For chlorine oxoacids, acid strength increases in the order $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$.
- (2) For oxoacids with the same central halogen, increasing number of terminal oxo groups strengthens the acid.
- (3) The increase in acid strength is associated with stronger electron withdrawal and greater stabilisation of the conjugate base.
- (4) HClO_4 contains chlorine in the highest common +7 oxidation state.
- (5) HClO contains chlorine in the +1 oxidation state.
- (6) Within analogous HOX acids, greater electronegativity of X generally strengthens the acid.
- (7) HOCl is therefore stronger than HOBr , which is stronger than HOI .
- (8) HClO is stronger than HClO_4 .
- (9) Increasing the oxidation state of chlorine from +1 to +7 weakens the corresponding oxoacid.
- (10) All halogen oxoacids contain fluorine in a positive oxidation state.
- (11) Oxidation state and number of oxo groups are distinct but related structural variables in oxoacid acidity.
- (12) The O-H bond is the ionisable acidic bond in the usual structural formulas HO-X(O)_n .
- (13) HClO_4 has chlorine in the -1 oxidation state.
- (14) The oxoacid formulas are consistent with one X-OH bond in each structure.
- (15) HOF proves that fluorine commonly exhibits +1 oxidation state.

Answer: _____

Q34. Interhalogens - reactivity, hydrolysis and uses [Count the correct statements.]

- (1) Most interhalogen compounds are volatile solids or liquids at 298 K, with ClF being a gas.
- (2) Their physical properties are generally intermediate between those of the constituent halogens.
- (3) Their melting and boiling points can be somewhat higher than simple expectation.
- (4) Interhalogens are generally more reactive than the halogens, except fluorine.
- (5) The enhanced reactivity is associated with weaker X-X' bonds than X-X bonds, except for the F-F anomaly.
- (6) Interhalogens undergo hydrolysis.
- (7) Hydrolysis gives halide ion derived from the smaller halogen.
- (8) The larger halogen appears in an oxyanion whose oxidation level depends on XX' , XX'_3 , XX'_5 or XX'_7 .
- (9) XX' hydrolysis gives a hypohalite-type product of the larger halogen.
- (10) XX'_3 , XX'_5 and XX'_7 correspond to halite, halate and perhalate-type oxyanions respectively.
- (11) XX'_3 compounds have bent T shapes.
- (12) XX'_5 compounds are square pyramidal.
- (13) IF_7 is pentagonal bipyramidal.
- (14) ClF_3 and BrF_3 are used as fluorinating agents in production of UF_6 for uranium enrichment.
- (15) $\text{U} + 3\text{ClF}_3 \rightarrow \text{UF}_6 + 3\text{ClF}$ is a fluorination reaction.

Answer: _____

Q35. BrF_3 VSEPR and geometry [Count the correct statements.]

- (1) Bromine has seven valence electrons in BrF_3 .
- (2) Three electron pairs form Br-F bonds.
- (3) Four electrons remain on bromine as two lone pairs.
- (4) BrF_3 therefore has three bond pairs and two lone pairs around the central atom.
- (5) Five electron pairs are arranged on a trigonal-bipyramidal electron-pair framework.
- (6) The two lone pairs occupy equatorial positions.
- (7) Equatorial lone pairs minimise lone-pair-lone-pair and lone-pair-bond-pair repulsions.
- (8) The axial fluorine atoms are bent slightly toward the equatorial fluorine.
- (9) The molecular shape is a slightly bent T.
- (10) Both lone pairs occupy axial positions.
- (11) BrF_3 has only one lone pair on bromine.
- (12) The electron-pair geometry is tetrahedral.
- (13) Lone-pair repulsions are greater than bond-pair-bond-pair repulsions in the VSEPR explanation.
- (14) The observed molecular geometry differs from the full five-position electron-domain geometry.
- (15) BrF_3 is square planar according to VSEPR.

Answer: _____

Q36. Group 17 reaction-equation mastery [Count the correct statements.]

- (1) $\text{F}_2 + 2\text{Cl}^- \rightarrow 2\text{F}^- + \text{Cl}_2$ is consistent with fluorine's oxidising power.
- (2) $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$.
- (3) $\text{Br}_2 + 2\text{I}^- \rightarrow 2\text{Br}^- + \text{I}_2$.
- (4) $2\text{F}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{F}^- + \text{O}_2$.
- (5) $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HOCl}$ represents chlorine hydrolysis.
- (6) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ prepares chlorine.
- (7) $4\text{HCl} + \text{O}_2 \rightarrow 2\text{Cl}_2 + 2\text{H}_2\text{O}$ is the Deacon reaction.
- (8) $8\text{NH}_3 + 3\text{Cl}_2 \rightarrow 6\text{NH}_4\text{Cl} + \text{N}_2$ occurs with excess ammonia.
- (9) $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$ occurs with hot concentrated alkali.
- (10) $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ occurs photochemically.
- (11) $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$ is the lower-temperature HCl preparation step.
- (12) $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$ gives white fumes.
- (13) $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$.
- (14) $\text{I}_2 + \text{Cl}_2 \rightarrow 2\text{ICl}$ is an interhalogen preparation.
- (15) $\text{Cl}_2 + 2\text{F}^- \rightarrow 2\text{Cl}^- + \text{F}_2$ occurs spontaneously in aqueous solution.

Answer: _____

Q37. Group 17 cumulative exceptions and high-yield traps [Count the correct statements.]

- (1) Cl has more negative electron gain enthalpy than F.
- (2) F_2 has lower bond dissociation enthalpy than Cl_2 .
- (3) HF has a much higher boiling point than HCl because of hydrogen bonding.
- (4) F_2 is nevertheless the strongest oxidising halogen.
- (5) Fluorine exhibits only -1 oxidation state in its compounds.
- (6) Iodine reaction with water is non-spontaneous in this comparison.
- (7) Bromine oxides are the least stable among the compared Cl, Br and I oxides.
- (8) Higher halogen oxides tend to be more stable than lower oxides.
- (9) Chlorine gives hypochlorite in cold dilute alkali but chlorate in hot concentrated alkali.
- (10) Excess ammonia and excess chlorine give different nitrogen-containing products.
- (11) Bleaching by chlorine is oxidative and permanent.
- (12) HCl is dried using concentrated H_2SO_4 .
- (13) Fluorine forms only one oxoacid, HOF .
- (14) Interhalogens are generally more reactive than the parent halogens except F_2 .

(15) All down-group trends in Group 17 are perfectly regular with no important fluorine anomalies.

Answer: _____

Q38. All colours, physical descriptions and visual-identification facts [Count the correct statements.]

- (1) F_2 is yellow and Cl_2 is greenish yellow.
- (2) Br_2 is red .
- (3) I_2 is violet .
- (4) Chlorine is a greenish-yellow gas and can be liquefied to a greenish-yellow liquid.
- (5) Hydrogen chloride is a colourless gas and forms a colourless liquid on liquefaction.
- (6) Solid HCl is described as white crystalline.
- (7) ClF is colourless.
- (8) BrF is pale brown.
- (9) $\alpha\text{-}ICl$ is ruby red and $\beta\text{-}ICl$ is brown red.
- (10) IBr is black.
- (11) ClF_3 is colourless, BrF_3 yellow green and IF_3 yellow.
- (12) ICl_3 is orange.
- (13) IF_5 and IF_7 are colourless .
- (14) BrF_3 is a violet gas.
- (15) IBr is a colourless liquid.

Answer: _____

Q39. All uses - halogen and interhalogen master coverage [Count the correct statements.]

- (1) ClO_2 is used for bleaching paper pulp and textiles.
- (2) ClO_2 is used in water treatment.
- (3) I_2O_5 is used in estimation of carbon monoxide.
- (4) Chlorine is used for bleaching wood pulp, cotton and textiles.
- (5) Chlorine is used in extraction of gold and platinum.
- (6) Chlorine is used in manufacture of dyes and drugs.
- (7) Chlorine is used in manufacture of CCl_4 , $CHCl_3$, DDT and refrigerants.
- (8) Chlorine is used to sterilise drinking water.
- (9) Chlorine is used to prepare phosgene, tear gas and mustard gas.
- (10) HCl is used to manufacture chlorine and NH_4Cl .
- (11) HCl is used to manufacture glucose from corn starch.
- (12) HCl is used in extraction of glue from bones and purification of bone black.
- (13) HCl is used in medicine and as a laboratory reagent.
- (14) Interhalogens can be used as non-aqueous solvents and as fluorinating agents.
- (15) ClF_3 and BrF_3 are used for UF_6 production in uranium enrichment.

Answer: _____

Q40. Atomic and physical data - Data mastery II [Count the correct statements.]

- (1) Melting points of F_2 , Cl_2 , Br_2 and I_2 are approximately 54.4, 172.0, 265.8 and 386.6 K.
- (2) Boiling points of F_2 , Cl_2 , Br_2 and I_2 are approximately 84.9, 239.0, 332.5 and 458.2 K.
- (3) The density values given are about 1.5, 1.66, 3.19 and 4.94 g cm⁻³ for F_2 , Cl_2 , Br_2 and I_2 under the specified conditions.
- (4) Hydration enthalpy magnitudes of F^- , Cl^- , Br^- and I^- are about 515, 381, 347 and 305 kJ mol⁻¹.
- (5) Standard reduction potentials E° are about 2.87, 1.36, 1.09 and 0.54 V for F_2/F^- , Cl_2/Cl^- , Br_2/Br^- and I_2/I^- .
- (6) F_2 has the lowest melting and boiling points among F_2 , Cl_2 , Br_2 and I_2 .
- (7) I_2 has the highest melting and boiling points among these four.
- (8) The standard reduction potential falls sharply from fluorine to iodine.
- (9) The hydration enthalpy magnitude of F^- greatly exceeds that of I^- .
- (10) The boiling point of Br_2 is lower than that of Cl_2 .
- (11) The standard reduction potential of I_2/I^- exceeds that of F_2/F^- .
- (12) Seaweeds can contain up to about 0.5% iodine.

(13) Chile saltpetre can contain up to about 0.2% sodium iodate.

(14) Sea water is described as containing about 25% $NaCl$ by mass.

(15) The physical data quantitatively support the increase of dispersion forces and phase-change temperatures down the group.

Answer: _____

Q41. Interhalogen table - complete fine-detail audit [Count the correct statements.]

- (1) IF is very unstable and has been detected spectroscopically.
- (2) $BrCl$ is an interhalogen compound of the XX' type.
- (3) ICl exists in two solid forms with different colours.
- (4) ICl_3 can dimerise as a chlorine-bridged species, I_2Cl_6 .
- (5) ClF_3 , BrF_3 , IF_3 and ICl_3 belong to the XX'_3 class.
- (6) IF_5 , BrF_5 and ClF_5 belong to the XX'_5 class.
- (7) IF_7 is the representative XX'_7 compound.
- (8) XX'_3 compounds are assigned bent T-shaped geometries.
- (9) XX'_5 compounds are square pyramidal.
- (10) IF_7 is pentagonal bipyramidal.
- (11) The interhalogens are covalent and diamagnetic.
- (12) Their physical properties are generally intermediate between those of the constituent halogens.
- (13) Their melting and boiling points may be somewhat higher than simple interpolation would suggest.
- (14) All interhalogens are ionic solids at 298 K.
- (15) IF_7 belongs to the XX'_5 class.

Answer: _____

Q42. Hydrogen chloride - quantitative and analytical fine details [Count the correct statements.]

- (1) The first laboratory step for HCl preparation is carried out near 420 K.
- (2) The second $NaHSO_4 + NaCl$ step is carried out near 823 K.
- (3) HCl has a boiling point near 189 K and freezing point near 159 K.
- (4) The dissociation constant quoted for HCl in water is of the order of 10^7 .
- (5) The very large dissociation constant is consistent with strong-acid behaviour in water.
- (6) Concentrated H_2SO_4 is suitable for drying HCl gas.
- (7) HCl gives dense white fumes of NH_4Cl with NH_3 .
- (8) Three volumes of concentrated HCl and one volume of concentrated HNO_3 form aqua regia.
- (9) Aqua regia dissolves gold and platinum by combined acidic, oxidising and complexing action.
- (10) Carbonates and hydrogencarbonates release CO_2 with HCl .
- (11) Sulphites release SO_2 with HCl .
- (12) Finely powdered iron gives $FeCl_2$ rather than $FeCl_3$ with HCl .
- (13) HCl is dried by passing it through concentrated $NaOH$.
- (14) The HCl/HNO_3 volume ratio in aqua regia is 1:3.
- (15) The combination of high water solubility and extensive ionisation explains the strong acidic action of aqueous HCl .

Answer: _____

Q43. ALL REACTION CONDITIONS AND OBSERVATIONS - Group 17 master audit [Count the correct statements.]

- (1) MnO_2 is heated with concentrated HCl to prepare chlorine in the laboratory.
- (2) Deacon's process oxidises HCl with atmospheric oxygen over $CuCl_2$ at about 723 K.
- (3) Chlorine is liberated at the anode during electrolysis of concentrated $NaCl$ solution.
- (4) With excess NH_3 , chlorine gives N_2 and NH_4Cl ; with excess Cl_2 , explosive NCl_3 is formed.

- (5) Cold dilute NaOH converts chlorine into chloride and hypochlorite.
- (6) Hot concentrated NaOH converts chlorine into chloride and chlorate.
- (7) Dry slaked lime reacts with chlorine to give bleaching powder.
- (8) Methane undergoes chlorination under ultraviolet light, whereas ethene gives an addition product with chlorine at room temperature.
- (9) The first NaCl-concentrated H_2SO_4 step for HCl preparation is carried out near 420 K.
- (10) The $\text{NaHSO}_4 + \text{NaCl}$ step for HCl preparation is carried out near 823 K.
- (11) Hydrogen chloride gas is dried by passing it through concentrated H_2SO_4 .
- (12) Aqua regia is prepared by mixing three volumes of concentrated HCl with one volume of concentrated HNO_3 .
- (13) Equal volumes of Cl_2 and F_2 at about 437 K give ClF , while excess F_2 at about 573 K gives ClF_3 .
- (14) Br_2 with 3F_2 gives BrF_3 under controlled conditions, while excess fluorine can give BrF_5 .
- (15) Cold dilute alkali and hot concentrated alkali give exactly the same chlorine disproportionation product mixture.

Answer: _____

Q44. Occurrence of halogens - complete natural- and abundance audit [Count the correct statements.]

- (1) Fluorine, chlorine, bromine, iodine and astatine are the elements of Group 17.
- (2) The name halogen comes from Greek roots meaning salt producer.
- (3) Halogens are highly reactive non-metallic elements.
- (4) Astatine is radioactive.
- (5) Fluorine and chlorine are fairly abundant, whereas bromine and iodine are less abundant.
- (6) Fluorine occurs mainly as insoluble fluorides such as fluor spar, CaF_2 .
- (7) Cryolite has the formula Na_3AlF_6 .
- (8) Fluoroapatite can be represented as $3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2$.
- (9) Small quantities of fluorine occur in soil, river water, plants, and in animal bones and teeth.
- (10) Sea water contains chlorides, bromides and iodides of sodium, potassium, magnesium and calcium.
- (11) Sea water is mainly sodium chloride solution, about 2.5% by mass.
- (12) Deposits of dried-up seas contain substances such as NaCl and carnallite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.
- (13) Certain seaweeds may contain up to about 0.5% iodine.
- (14) Chile saltpetre can contain up to about 0.2% sodium iodate.
- (15) Iodine is more abundant than chlorine in sea water.

Answer: _____

Q45. Halogen table - exact atomic, thermochemical and physical data [Count the correct statements.]

- (1) Atomic numbers of F, Cl, Br, I and At are 9, 17, 35, 53 and 85.
- (2) Approximate atomic masses are 19.00, 35.45, 79.90, 126.90 and 210 g mol^{-1} .
- (3) Covalent radii of F, Cl, Br and I are approximately 64, 99, 114 and 133 pm.
- (4) X^- ionic radii of F^- , Cl^- , Br^- and I^- are approximately 133, 184, 196 and 220 pm.
- (5) First ionisation enthalpies of F, Cl, Br and I are approximately 1680, 1256, 1142 and 1008 kJ mol^{-1} .
- (6) Electron gain enthalpies of F, Cl, Br and I are about -333, -349, -325 and -296 kJ mol^{-1} .
- (7) Electronegativities are approximately 4.0, 3.2, 3.0, 2.7 and 2.2 for F, Cl, Br, I and At.
- (8) Magnitudes of hydration enthalpy for F^- , Cl^- , Br^- and I^- decrease roughly as 515, 381, 347 and 305 kJ mol^{-1} .

- (9) Melting points of F_2 , Cl_2 , Br_2 and I_2 are about 54.4, 172.0, 265.8 and 386.6 K.
- (10) Boiling points of F_2 , Cl_2 , Br_2 and I_2 are about 84.9, 239.0, 332.5 and 458.2 K.
- (11) X-X bond distances for F_2 , Cl_2 , Br_2 and I_2 are about 143, 199, 228 and 266 pm.
- (12) Bond dissociation enthalpies are about 158.8, 242.6, 192.8 and 151.1 kJ mol^{-1} for F_2 , Cl_2 , Br_2 and I_2 .
- (13) Standard reduction potentials for F_2/F^- , Cl_2/Cl^- , Br_2/Br^- and I_2/I^- are approximately 2.87, 1.36, 1.09 and 0.54 V.
- (14) Electron gain enthalpy of fluorine is more negative than that of chlorine.
- (15) Among F_2 , Cl_2 , Br_2 and I_2 , the Cl-Cl bond has the smallest bond dissociation enthalpy.

Answer: _____

Q46. Halogen physical properties - colours, states, solubility and bond anomaly [Count the correct statements.]

- (1) All Group 17 elements have seven electrons in the outermost shell with general configuration ns^2np^5 .
- (2) Halogens have the smallest atomic radii in their respective periods because of high effective nuclear charge.
- (3) Atomic and ionic radii increase from fluorine to iodine.
- (4) Ionisation enthalpy is high for halogens and decreases down the group.
- (5) Halogens have very negative electron gain enthalpies in their respective periods.
- (6) Electron gain enthalpy generally becomes less negative down the group, but fluorine is less negative than chlorine.
- (7) The relatively small 2p shell of fluorine leads to strong interelectronic repulsion for an incoming electron.
- (8) Electronegativity decreases down Group 17 and fluorine is the most electronegative element.
- (9) F_2 and Cl_2 are gases, Br_2 is a liquid and I_2 is a solid under ordinary conditions.
- (10) Melting and boiling points increase steadily down the group.
- (11) All halogens are coloured because visible radiation can excite outer electrons.
- (12) F_2 is yellow, Cl_2 greenish-yellow, Br_2 red and I_2 violet.
- (13) Fluorine and chlorine react with water, whereas bromine and iodine are only sparingly soluble in water.
- (14) Bromine and iodine dissolve in organic solvents such as chloroform, carbon tetrachloride, carbon disulphide and hydrocarbons to give coloured solutions.
- (15) The F-F bond is stronger than the Cl-Cl bond.

Answer: _____

Q47. Fluorine anomaly and hydrogen halides - complete numerical trend audit [Count the correct statements.]

- (1) Anomalous behaviour of fluorine is associated with small size, highest electronegativity, low F-F bond dissociation enthalpy and absence of valence d orbitals.
- (2) Most reactions of fluorine are exothermic because strong bonds are formed with other elements.
- (3) Fluorine forms only one oxoacid in the cited series.
- (4) HF is a liquid with boiling point about 293 K because of strong hydrogen bonding.
- (5) The other common hydrogen halides are gases under ordinary conditions.
- (6) Affinity of halogens for hydrogen decreases from fluorine to iodine.
- (7) Aqueous acid strength follows $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$.
- (8) Thermal stability follows $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$.
- (9) H-X bond lengths for HF, HCl, HBr and HI are about 91.7, 127.4, 141.4 and 160.9 pm.
- (10) H-X bond dissociation enthalpies are about 574, 432, 363 and 295 kJ mol^{-1} respectively.

(11) Approximate pK_a values listed are 3.2, -7.0, -9.5 and -10.0 for HF, HCl, HBr and HI.

(12) HF has a melting point around 190 K and boiling point around 293 K.

(13) HCl has a boiling point near 189 K.

(14) HI is a stronger aqueous acid than HBr.

(15) HF has the weakest H-X bond among the hydrogen halides.

Answer: _____

Q48. Halogen oxides - complete stability, oxidising action and special uses [Count the correct statements.]

(1) Fluorine forms OF_2 and O_2F_2 .

(2) Of these two oxygen fluorides, only OF_2 is thermally stable at 298 K.

(3) OF_2 and O_2F_2 are described as oxygen fluorides because fluorine is more electronegative than oxygen.

(4) Both OF_2 and O_2F_2 are strong fluorinating agents.

(5) O_2F_2 can oxidise plutonium to PuF_6 , a reaction used in removal of plutonium from spent nuclear fuel.

(6) Chlorine, bromine and iodine form oxides in which halogen oxidation states range from +1 to +7.

(7) The general stability order quoted for halogen oxides is $I > Cl > Br$.

(8) Higher oxides of halogens tend to be more stable than lower oxides.

(9) Cl_2O , ClO_2 , Cl_2O_6 and Cl_2O_7 are highly reactive oxidising agents and can be explosive.

(10) ClO_2 is used for bleaching paper pulp and textiles and in water treatment.

(11) Br_2O , BrO_2 and BrO_3 are among the least stable halogen oxides, exist only at low temperatures and are strong oxidants.

(12) I_2O_4 , I_2O_5 and I_2O_7 are insoluble solids that decompose on heating.

(13) I_2O_5 is used in estimation of carbon monoxide because it is a good oxidising agent.

(14) Bromine oxides are the most stable oxides among Cl, Br and I.

(15) OF_2 is more appropriately described as an oxygen fluoride than as an oxide of fluorine.

Answer: _____

Q49. Metal halides and interhalogen principles - covalency and composition [Count the correct statements.]

(1) Bromine reacts with magnesium to form $MgBr_2$.

(2) For a monovalent metal M, ionic character decreases in the order $MF > MCl > MBr > MI$.

(3) When a metal has more than one oxidation state, the halide in the higher oxidation state is generally more covalent.

(4) $SnCl_4$ is more covalent than $SnCl_2$.

(5) $PbCl_4$ is more covalent than $PbCl_2$.

(6) $SbCl_5$ is more covalent than $SbCl_3$.

(7) UF_6 is more covalent than UF_4 .

(8) Interhalogen compounds occur in types XX' , XX'_3 , XX'_5 and XX'_7 .

(9) In these formulas X is generally the larger halogen and X' the smaller halogen.

(10) X is more electropositive than X' in the usual interhalogen notation.

(11) As the radius ratio X/X' becomes larger, the maximum number of atoms that can be accommodated can increase.

(12) IF_7 has the maximum atom count among the common interhalogen types listed.

(13) Interhalogen compounds are predominantly ionic salts.

(14) Interhalogens listed are covalent molecules and diamagnetic.

(15) Interhalogen compounds are generally more reactive than the constituent halogens except fluorine.

Answer: _____

Q50. Chlorine - discovery, laboratory preparation, manufacture and physical data [Count the correct statements.]

(1) Chlorine was discovered by Scheele in 1774 by the action of HCl on MnO_2 .

(2) Davy established its elementary nature in 1810 and the name chlorine reflects its greenish-yellow colour.

(3) $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$ is a laboratory preparation reaction.

(4) A mixture of common salt and concentrated H_2SO_4 can be used in place of ready-made HCl in the MnO_2 method.

(5) An overall preparation equation is $4NaCl + MnO_2 + 4H_2SO_4 \rightarrow MnCl_2 + 4NaHSO_4 + 2H_2O + Cl_2$.

(6) $KMnO_4$ oxidises HCl according to $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$.

(7) In the Deacon process HCl is oxidised by atmospheric oxygen using $CuCl_2$ catalyst at about 723 K.

(8) Electrolysis of concentrated NaCl solution liberates chlorine at the anode.

(9) Chlorine is also obtained as a by-product in several chemical industries.

(10) Chlorine is a greenish-yellow gas with pungent and suffocating odour.

(11) Chlorine is heavier than air.

(12) It can be liquefied to a greenish-yellow liquid boiling at about 239 K.

(13) Chlorine is soluble in water.

(14) Elemental chlorine is colourless.

(15) In brine electrolysis, chlorine is liberated at the cathode.

Answer: _____

Q51. Chlorine - element, ammonia, alkali and bleaching-powder reaction set [Count the correct statements.]

(1) $2Al + 3Cl_2 \rightarrow 2AlCl_3$ is a reaction of chlorine with aluminium.

(2) $P_4 + 6Cl_2 \rightarrow 4PCl_3$ is a cited reaction of chlorine with phosphorus.

(3) $2Na + Cl_2 \rightarrow 2NaCl$ represents formation of sodium chloride.

(4) $S_8 + 4Cl_2 \rightarrow 4S_2Cl_2$ is a cited sulphur-chlorine reaction.

(5) $2Fe + 3Cl_2 \rightarrow 2FeCl_3$ shows formation of ferric chloride.

(6) $H_2 + Cl_2 \rightarrow 2HCl$ illustrates the strong affinity of chlorine for hydrogen.

(7) $H_2S + Cl_2 \rightarrow 2HCl + S$ shows oxidation of hydrogen sulphide by chlorine.

(8) $C_{10}H_{16} + 8Cl_2 \rightarrow 16HCl + 10C$ is a cited reaction of chlorine with a hydrogen-containing compound.

(9) With excess NH_3 , $8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$.

(10) With excess chlorine, $NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$, and NCl_3 is explosive.

(11) Cold dilute NaOH converts chlorine into chloride and hypochlorite.

(12) Hot concentrated NaOH converts chlorine into chloride and chlorate.

(13) With dry slaked lime, chlorine gives a mixture containing hypochlorite and chloride species used as bleaching powder.

(14) Bleaching powder is represented as $Ca(OCl)_2 \cdot CaCl_2 \cdot Ca(OH)_2 \cdot 2H_2O$.

(15) Chlorine reacts with ethene only by substitution and does not add across the double bond.

Answer: _____

Q52. Chlorine water - hydrocarbon reactions, oxidation equations and complete uses [Count the correct statements.]

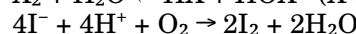
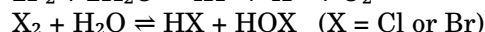
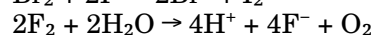
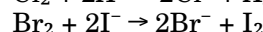
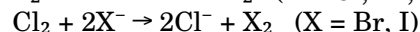
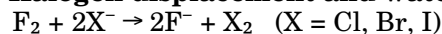
(1) Under UV light, $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ is a substitution reaction.

(2) At room temperature, ethene adds chlorine to give 1,2-dichloroethane.

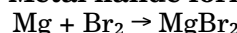
(3) Chlorine water loses its yellow colour on standing because HCl and HOCl are formed.

REACTION BANK — GROUP 17

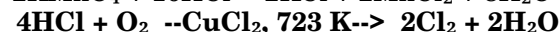
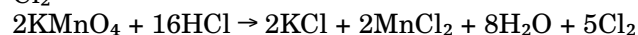
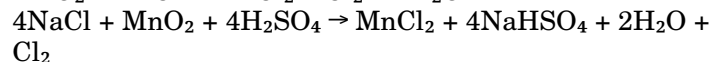
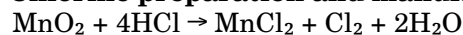
Halogen displacement and water reactions



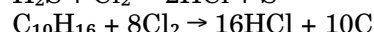
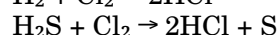
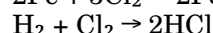
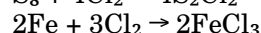
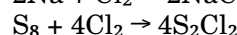
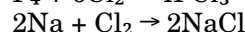
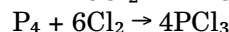
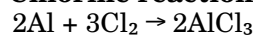
Metal halide formation



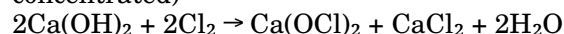
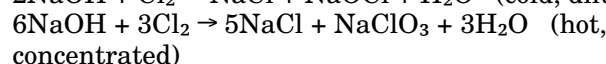
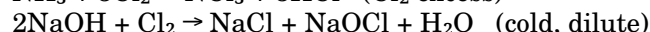
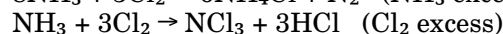
Chlorine preparation and manufacture



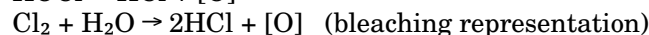
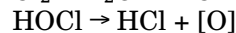
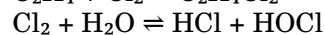
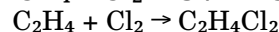
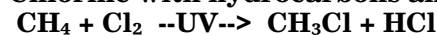
Chlorine reactions with elements and compounds



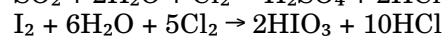
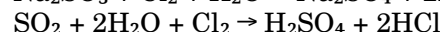
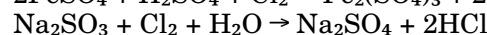
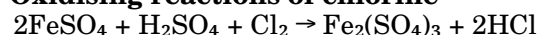
Chlorine with ammonia, alkali and lime



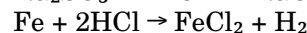
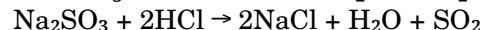
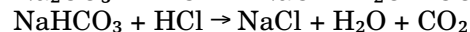
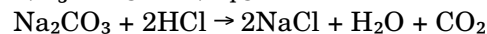
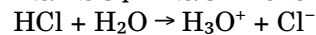
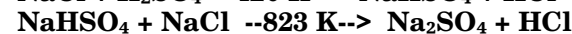
Chlorine with hydrocarbons and water



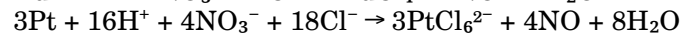
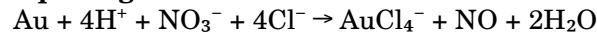
Oxidising reactions of chlorine



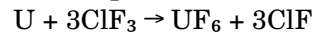
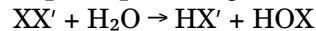
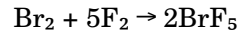
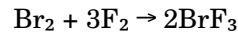
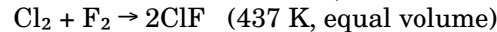
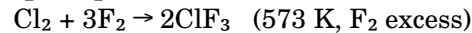
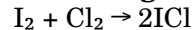
Hydrogen chloride



Aqua regia reactions as written in ionic form



Interhalogen preparation and use



- (4) HOCl can furnish active oxygen responsible for oxidising and bleaching action.
- (5) Chlorine oxidises Fe(II) to Fe(III).
- (6) Chlorine oxidises sulphite to sulphate.
- (7) $\text{SO}_2 + 2\text{H}_2\text{O} + \text{Cl}_2 \rightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$ is a cited oxidation reaction.
- (8) $\text{I}_2 + 6\text{H}_2\text{O} + 5\text{Cl}_2 \rightarrow 2\text{HIO}_3 + 10\text{HCl}$ shows oxidation of iodine to iodic acid.
- (9) Bleaching action of chlorine is an oxidation process.
- (10) Bleaching by chlorine is permanent.
- (11) Chlorine is used in bleaching wood pulp, cotton and textiles.
- (12) Chlorine is used in extraction of gold and platinum.
- (13) It is used in manufacture of dyes, drugs and compounds such as CCl_4 , CHCl_3 , DDT and refrigerants.
- (14) It is used for sterilising drinking water.
- (15) Chlorine is used in preparation of poisonous gases such as phosgene, tear gas and mustard gas.

Answer: _____

Q53. Halogen oxoacids and interhalogen preparation - exact formula and reaction audit [Count the correct statements.]

- (1) Because of its very high electronegativity and small size, fluorine forms only one oxoacid, HOF.
- (2) Many oxoacids of the other halogens cannot be isolated pure and are stable mainly in aqueous solution or as salts.
- (3) HOF, HOCl, HOBr and HOI are the +1 hypohalous acids listed.
- (4) HClO_2 is the +3 chlorous acid listed in the series.
- (5) HClO_3 , HBrO_3 and HIO_3 correspond to the +5 halic acids.
- (6) HClO_4 , HBrO_4 and HIO_4 correspond to the +7 perhalic acids in the table.
- (7) In the oxoacid structures the halogen is central and is bonded through an X-OH group; additional X=O bonds occur in higher members.
- (8) Equimolar I_2 and Cl_2 combine to form ICl .
- (9) With excess fluorine at about 573 K, chlorine forms ClF_3 .
- (10) Equal volumes of Cl_2 and F_2 at about 437 K form ClF .
- (11) I_2 with excess chlorine can form ICl_3 .
- (12) Br_2 with three equivalents of F_2 can form BrF_3 .
- (13) Br_2 with five equivalents of F_2 can form BrF_5 .
- (14) Interhalogen compounds are covalent and diamagnetic; most are volatile solids or liquids, while ClF is a gas.
- (15) On hydrolysis, the smaller halogen gives halide while the larger halogen gives an oxyanion whose oxidation level depends on XX' , XX'_3 , XX'_5 or XX'_7 stoichiometry.

Answer: _____

PART II — GROUP 17

Q28: 14	Q29: 12	Q30: 11	Q31: 13	Q32: 11	Q33: 10
Q34: 15	Q35: 11	Q36: 14	Q37: 14	Q38: 13	Q39: 15
Q40: 12	Q41: 13	Q42: 13	Q43: 14	Q44: 14	Q45: 13
Q46: 14	Q47: 14	Q48: 14	Q49: 14	Q50: 13	Q51: 14
Q52: 15	Q53: 15				